

Section 4

Systems Programming ECE3502

SIC/XE Machine Architecture

Memory formats

1 megabytes (2^{20} bytes)

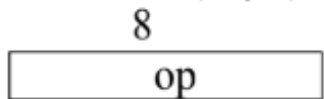
Registers

There are nine registers; each register is 24 bits in length except floating point register

name	width (bits)	description	Number
A	24	accumulator	0
X	24	index register	1
L	24	linkage register	2
B	24	base register	3
S	24	general purpose register	4
T	24	general purpose register	5
F	48	floating point register	6
PC	24	program counter	8
SW	24	Status Word	9

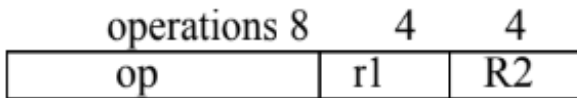
Instruction formats

Format 1 (1 byte): no memory ref.



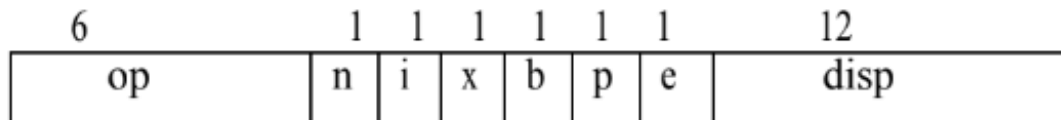
Example: TIO

Format 2 (2 bytes): no memory ref. / register-to-register



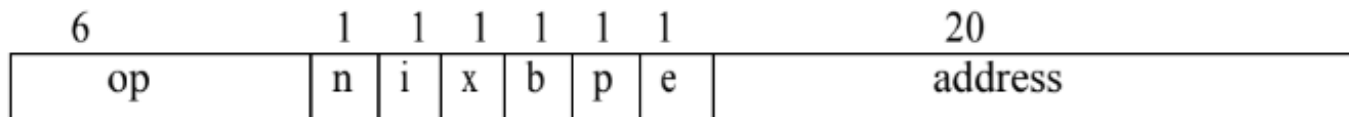
Example: ADDR S,T

Format 3 (3 bytes):



Example: LDA #3

Format 4 (4 bytes):



Example: +JSUB RDREC

- Bit **e** is used to distinguish between format 3 and 4 (**e=0** means Format 3, **e=1** means Format 4). [“+” is added to the instruction]

SIC/XE Addressing Modes

Mode	Indication	Target address calculation
Base relative	b=1,p=0	TA=(B)+disp ($0 \leq \text{disp} \leq 4095$)
Program-counter relative	b=0,p=1	TA=(PC)+disp ($-2048 \leq \text{disp} \leq 2047$)
Direct addressing	b=0,p=0	TA=disp

- Any of these addressing modes can also be combined with indexed addressing if bit x is set to 1 the term (x) is added in the target address

Immediate “#”	i=1,n=0	Value= TA
Indirect Addressing “@”	i=0,n=1	Value=((TA))
Simple	i=1,n=1	Value= (TA)

- Indexing cannot be used with immediate or indirect addressing mode.
- If bits n and i are both 0, then bits b, p, and e are considered to be part of the address field of the instruction (rather than flags indicating addressing mode). This makes Instruction Format 3 identical to the format used on the standard version of SIC, providing the desired compatibility

Sheet3, Q1)

Given the following memory structure calculate the target address and find the value loaded into register A for the following machine instructions:

1. 032600

opcode	n	i	x	b	p	e	Disp.
0000 00	1	1	0	0	1	0	0110 0000 0000

1 Get address: (b,p,x)

TA=disp+ (PC)

TA=600+3000

2 Use address: (n,i)

TA=3600

Value=(TA)= (3600)

Value=103000

2. 03C300

opcode	n	i	x	b	p	e	Disp.
0000 00	1	1	1	1	0	0	0011 0000 0000

1 Get address: (b,p,x)

TA=disp+ (B)+ (x)

TA=300+6000+90

2 Use address: (n,i)

TA=6390

Value=(TA)= (6390)

Value=00C303

Memory

...	...
3030	003600
...	...
3600	103000
...	...
6390	00C303
...	...
C303	003030
...	...

(a)

(B) = 006000

(PC) = 003000

(X) = 000090

Sheet3, Q1)

3. 022030

opcode	n	i	x	b	p	e	Disp.
0000 00	1	0	0	0	1	0	0000 0011 0000

1 Get address: (b,p,x)

TA=disp+ (PC)

TA=030+3000

2 Use address: (n,i)

TA=3030

Value=((TA))= ((3030))

Value= (3600) = 103000

4. 010030

opcode	n	i	x	b	p	e	Disp.
0000 00	0	1	0	0	0	0	0000 0011 0000

1 Get address: (b,p,x)

TA=disp

TA= 030

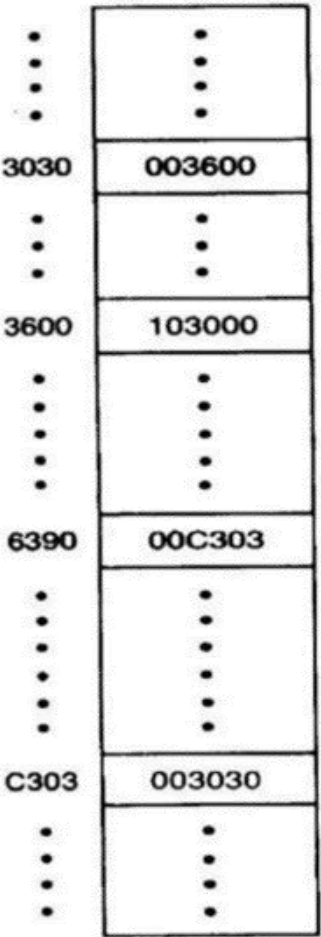
2 Use address: (n,i)

TA=030

Value=TA= 030

Value=000030

Memory



(B) = 006000
(PC) = 003000
(X) = 000090

(a)

Sheet3, Q1)

5. 0310C303

opcode	n	i	x	b	p	e	Add.
0000 00	1	1	0	0	0	1	0000 1100 0011 0000 0011

- 1 Get address: (b,p,x)

TA=address

TA=0C303
- 2 Use address: (n,i)

TA= 0C303

Value=(TA)= (0C303)

Value=003030

6. 003600

opcode	n	i	x	b	p	e	Add.
0000 00	0	0	0	0	1	1	0110 0000 0000

Opcode(8) x Address(15)

0000 0000	0	0110110000000000
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- 1 Get address: (b,p,x)

TA=address

TA=3600
- 2 Use address: (n,i)

TA= 3600

Value=(TA)= (3600)

Value=103000

